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Telecommunication Support for Rural Women With Diabetes

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PURPOSE

The Women to Women Diabetes Project tested the use of telecommunication technology to deliver diabetes education and social support to rural women with diabetes. The aims were to (1) test the effects of the computer intervention in providing support, information, and education on selected outcomes, and (2) evaluate the women's attitudes toward and satisfaction with the intervention and the support provided.

METHODS

Thirty women were randomized into computer and noncomputer groups and participated for 10 months. For 5 months, one group participated in a self-help support and educational group via the computer; the other group continued to use their usual modes of support and communication. Psychosocial well-being scales were administered and attitudes were surveyed.

RESULTS

Improving health and higher educational levels positively influenced measures of social support and quality of life. Women who were married or who reported greater support had higher scores on the Personal Resource Questionnaire. The women expressed positive effects of the computer-based support group on their lives.

CONCLUSIONS

The intervention was enthusiastically accepted, and could be conducted effectively in isolated rural areas.

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Rurality can be a complicating factor for diabetes educators and their clients in the process of trying to deliver information about diabetes and provide support in managing the disease. One method of overcoming this difficulty is using computer telecommunication. This study of rural women with diabetes focused on the intentional use of computer telecommunication modalities to teach diabetes self-management and to form support networks that could span many miles and be functional regardless of external conditions such as weather. The aims of the Women to Women Diabetes Project (WTWDP) were to (1) test the effects of using telecommunications technology to provide support, information, and education on selected outcomes (physical and psychosocial) for rural women with diabetes; and (2) evaluate the women's attitudes toward and satisfaction with the various components of the intervention (eg, electronic classroom sessions, contact with a certified diabetes educator) and the support offered by other rural women with diabetes.

BACKGROUND

This article focuses on the role of social support in successfully living with diabetes, barriers to providing information and support to women with diabetes living in rural areas, and the use of telecommunication to overcome these barriers. The role of the certified diabetes educator (CDE) centers around providing information to clients about which they can make self-care decisions. Providing information is also one of the parts of social support as defined by Weiss.[1] Of particular importance is the notion that control of a chronic illness such as diabetes can be positively influenced by social support.[2] As a result, teaching in a supportive, interactive environment in which a person with diabetes can get information from a professional and process it with other people who have diabetes should result in positive measures of glycemic control, psychological adaptation, and quality of life.

While social support has been positively linked to both glycemic control and quality of life for people with diabetes,[3] those living in rural areas may not have access to supportive interventions.[4] In many rural areas, interventions that require regular face-to-face interactions are not feasible for a number of reasons. Rural residents are characterized by self-reliance and independence, rely heavily on an informal network of support,[5] and may not be accepting of externally provided support. Moreover, rural areas have relatively few healthcare resources and lack adequate transportation systems.[6] The distances to sources of health care may be formidable. In one Montana study,[7] the average distance traveled for appointments with an oncologist was 172 miles round trip. In large, sparsely populated areas such as Montana, it is impossible to provide some of the typical established support programs. For example, the *Encore Program* sponsored by the YWCA for breast cancer survivors is not available in Montana because very few YWCA facilities exist in the state. In rural areas, distance, travel time, weather conditions, and transportation requirements potentially prohibit establishing and maintaining adequate communication between the CDE and people with diabetes, which, in turn, makes it difficult to facilitate the necessary social support.

Telecommunication promises to be a viable alternative for supportive interventions in rural areas. Telephones have been used for some time to improve support and communication in both rural and urban areas.[8] Unfortunately, using telephones for extended periods can be impractical and time-intensive for both the professional and the client. Conversely, individuals can use computer telecommunication at their convenience, allowing for asynchronous but interactive communication. Home computers are being used for a variety of tasks including education, home monitoring of physical parameters, home budgets, entertainment, and business. It is estimated that 35% of all American households have a computer.[9] In a survey[10] of Montana farmers and ranchers, 20% reported using personal computers to manage their operations. Telecommunication systems are also used to expand educational programs, link healthcare providers, and conduct business.[11]

Computer technology is being used more frequently to impact diabetes care. Computers are being used to chart patient care, facilitate blood glucose monitoring, and manage databases for reporting aggregate population data. Diabetes self-management training has been improved by the use of computers, first, by making it possible to easily individualize material for a specific patient, and, second, by the ease of updating records for use in follow-up training.

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Computers also have the potential of improving diabetes care by providing the necessary components of a therapeutic or support group.[12] Telecommunication systems have been used successfully to provide support for different urban groups, such as care providers of chronically ill patients,[13] individuals with AIDS, women with breast cancer,[14] and those trying to stop smoking.[15] Those in computer support groups for caregivers and individuals with AIDS discussed disease information, available community resources, behavior management, and coping skills. Women with breast cancer "talked" about past and ongoing treatment, physical and emotional symptoms, diagnostic tests, and interactions with medical personnel.[16] A computer intervention group for urban women with breast cancer reduced depression and isolation.[14] Despite being less familiar with computers, older caregivers were able with limited assistance to learn to use computers,[17] resulting in enhanced self-esteem, a reduction in feelings of loneliness, and facilitation of other methods of communication.

Education and social support have been known to increase the quality of self-management by people with diabetes. In addition, computer telecommunication technology has been shown to effectively facilitate education and social support in diabetes.[18] This study was significant in that it used computer telecommunication to offer education and social support to rural women with diabetes, as well as to measure its effects.

METHODS

Thirty women with diabetes who met the inclusion criteria participated in the study over 10 months. The implementation of the study included recruitment, screening, gaining consents, randomization into groups, data collection via telephone interviews and mail questionnaires, and administering the intervention. The study was reviewed and approved by the Institutional Review Board of Montana State University.

Inclusion Criteria

Women who participated in the Women to Women Diabetes Project had to meet the following study inclusion criteria: have diabetes (type 1 or 2), be between the ages 35 and 60 years, be able to read and write English, and possess the sight and dexterity to use a computer (but not necessarily be computer literate). In addition, participants were required to have a telephone in their homes and live at least 25 miles outside the 6 major cities of Montana.

Practical Application

People with diabetes are more apt to engage in self-care behaviors when their social networks provide practical and emotional support. Because much of diabetes self-care takes place in a social context, evaluating and understanding social influences are often critical to facilitating effective self-management. Diabetes educators who work with people with diabetes who live in rural areas, however, may need to use a different approach to providing support. For many rural residents, access to diabetes education and care is difficult because of geographic isolation and limited numbers of specialized healthcare providers. The need to travel long distances adds a financial, physical, and emotional burden to an already stressed patient and family. In addition, lack of healthcare providers in some rural areas makes it difficult for diabetes educators to network with patients and other colleagues for brainstorming, support, and collaboration. Smith and Weinert have conducted a study evaluating the effectiveness of a telecommunication system in providing the necessary components of a therapeutic support group to rural women who have diabetes. The women in this study expressed affirmative comments regarding use of the computer technology in communicating with other women and the diabetes educator. This study provided evidence that use of telecommunication through computers may be a viable and creative approach to providing social support. Further research could examine the effectiveness of such technology in delivering diabetes education to various rural populations.

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Recruitment and Screening

Recruitment began by sending letters that explained the WTWDP to CDEs throughout Montana. Informational flyers were enclosed for distribution to appropriate women in their practices. Assistance also was provided by the American Diabetes Association, Montana Affiliate (ADA-MT), by screening its mailing list to identify potential participants and mailing the flyers from its office to maintain confidentiality.

Fifty women responded to the recruitment efforts by contacting the project via the toll-free number. Each woman who responded was called by a CDE and a screening interview was conducted. The screening interview included questions that would verify whether the person met the inclusion criteria, collect demographic information, inquire about computer literacy, and assess for the availability of a computer and/or online capabilities. During the screening interview, the CDE explained the project in detail and emphasized the fact that agreeing to participate meant agreeing to be placed in either the computer or noncomputer group.

Consent and Randomization

The first 30 women who completed the screening interview, met the criteria, and were interested in the project were included in the data pool. Those who were not in the data pool were asked for permission to keep their names on the project roster so they could be contacted for later study cohorts. A consent to participate was sent to each of the women selected and returned to the project office. The 30 women were randomized into two groups (computer vs noncomputer) of 15 each. After the randomization was completed, the women were sent another consent form that was specific to the group for which they had been randomly assigned.

Data Collection

Data were collected through telephone interviews, mail questionnaires, and compilation of computer-use information. First, telephone interviews were conducted at baseline, 2½ months, 5 months, 7½ months, and 10 months and included a change in health status scale, a sources of support scale, and a self-report of the date and level of the most recent HbA_{1c}. The Montana State University Rurality Index questions were incorporated into the demographic information collected from the participants in the telephone screening interview to objectively measure the impact on rural dwellers of living long distances from healthcare resources.

The second means of collecting data was using mail questionnaires, which were sent out with an instructional letter, at baseline, 2½ months, 5 months, 7½ months, and 10 months. This mailing included the Personal Resource Questionnaire (PRQ),[19] which is a measure of global social support; the Quality of Life (QoL) Index[20]; the Social Readjustment Rating Scale,[21] which is an assessment of major life stresses; and the Psychosocial Adaptation to Illness Scale (PAIS).[22]

The computer group had access to the support intervention for the first 5 months. In the sixth month, an attitude survey was mailed to the women in the computer group to assess their attitudes about the impact of the 5-month computer intervention on their lives. Questions were designed to address the issues regarding use of the computer, degree of comfort with the program, and need for assistance in using the program. Two open-ended questions were included to elicit ideas for refining the program and assess the women's overall perspective on the experience of participating in the project.

The third source of data was the compilation of statistics on the use of the computer intervention that were collected by the software used for the intervention. The data collected included the number of times logged onto the system and the length of the sessions.

Intervention

The 5-month computer intervention involved installing the computers and/or computer software in each of the women's homes and instructing them in its use; providing each woman with a notebook of health information regarding women's health in general and specific diabetes information (with special attention to its effect on women); and implementing the software system (FirstClass) for the computer intervention. During the computer intervention phase, the noncomputer group was given a hard copy of all the materials that the computer group was given (health information notebook and computer-generated information); they did not receive a computer and computer training.

The FirstClass software system, version 5.506 (SoftArc, Inc, Markham, Ontario, Canada), is a multiplatform client/server communication system used by the Montana State University Burns Telecommunication Center (BTC). The BTC furnished technical support for the software aspects of the WTWDP.

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To install the computers or software, a nurse CDE made appointments with each of the women and traveled to their homes. A laptop computer that contained the FirstClass software was loaned to those who did not have computers. For those who had a compatible computer, the FirstClass software was installed. Everyone in the computer group was trained in the use of the software and how to log on and connect to the main system. The FirstClass software was accessed directly via a toll-free modem hookup, whether the woman had her own computer or was using a laptop. At the first visit, issues of confidentiality and support group participation were reviewed and the protocol of the project was explained.

The FirstClass software was adapted for this study and consisted of 4 components: "Conversation," "Mailbox," "HealthChat," and "Resource Rack." Women had 24-hour access to the system and all computer exchanges were asynchronous.

In the "Conversation" area, the women were encouraged to converse with each other about anything that interested them in any areas of their lives. This area functioned much like a support group. The comments were open and available to be read by any of the women and the study team. The exchanges were monitored daily by the CDE nurse monitor. However, she did not actively participate in the exchanges unless directly invited by the women or there was a need to redirect inappropriate participation by group members.

The "Mailbox" area provided an opportunity for the women to "speak" privately to each other or to the nurse monitor. These private messages were seen only by the 2 participants.

The "HealthChat" area, the most structured of the components, was designed as the platform for diabetes education, much like an electronic classroom. This area was open for discussions by all members of the group as was the "Conversation" area. Questions specific to dealing with diabetes were encouraged and articles from the health information notebook were discussed. The CDE took an active role in the "HealthChat" area as opposed to her passive role in the "Conversation" area.

Table 1.

Raw Mean Scores for the Psychosocial Assessments

Assessment	Computer Group	Noncomputer Group
PAIS	77.79	80.24
PRQ	121.76	128.53
QoL	17.1755	17.9014

PAIS=Psychosocial Adjustment to Illness Scale.
PRQ=Personal Resource Questionnaire.
QoL=Quality of Life Scale.

"Resource Rack" was the fourth component and functioned as a bulletin board on which the project team posted items of interest to people with diabetes, such as the dates and times of the ADA-MT conference. This area was set up as a read-only feature and did not allow participants to post items. In addition to computer access to the study team that was available to the computer group, the women in both groups were encouraged to use the toll-free number to contact the members of the team if they had any questions or concerns about the project.

RESULTS

Demographics

The mean age of the women who participated in the study was 46.7 years. Most (80%) of the participants had type 2 diabetes, 80% reported being currently married, and 60% were employed. The participant considered the most rural lived 90 miles from the nearest source of health care. Two participants had computers that could accommodate the FirstClass software; the rest of the women in the computer group were loaned a laptop computer.

Telephone Interviews and Mail Questionnaires

The women who had access to the intervention positively evaluated its effect on their psychosocial and physical health. The differences between the mean scores of the computer and noncomputer groups for each of the psychosocial assessments scores are shown in Table 1. Higher scores on the QoL scale are associated with perceptions of a higher quality of life. Higher scores on the PRQ indicate higher levels of social support. Lower PAIS scores show better adjustment to illness. Because the sample size of this cohort was small, the analyses could not yield definitive results. Nonetheless, the analyses can give some clues about how the computer intervention and other explanatory variables might influence the well-being of women with diabetes.

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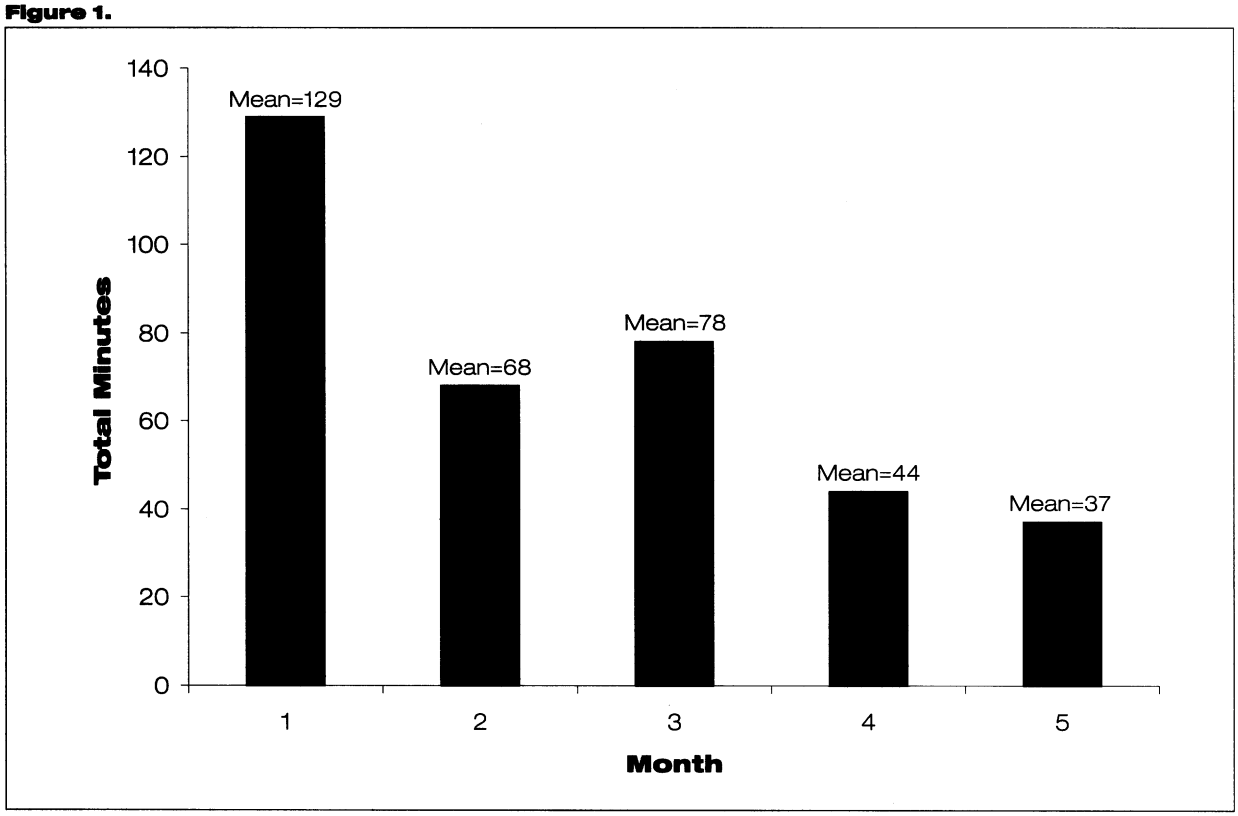
Figure 1.



Mean minutes logged on per month; total mean=63.80.

The raw data were adjusted for 7 covariates: Social Readjustment Rating Scale scores, the MSU Rurality Index, marital status, employment status, self-reported support currently received from family and others, years of education, and major health changes occurring during the study. After these adjustments, the Quality of Life score was higher for women (in both groups) who had more years of education and whose health was improving. There was no apparent difference between the computer and noncomputer groups when adjusted for these covariates. Similarly, there were no significant differences between the groups when adjusted for the covariates for the PRQ. Improving health and higher educational levels positively influenced social support scores and quality of life scores. In addition, women who were married or who reported greater support during the telephone interviews had higher PRQ scores.

Although statistical modeling of the PAIS continues, preliminary analysis indicated that employment may have a significant effect on the women's adjustment to their illness. Overall, the participants in both groups tended to feel better adjusted to their disease with the passage of time. The computer group had somewhat better PAIS scores, although the differences were not statistically significant.

The attempt to use HbA_{1c} as a physical measure of intervention effectiveness was subject to the hazards of self-reporting of laboratory data. Only 2 women in the computer group and 3 women in the noncomputer group had access to HbA_{1c} values that would reflect blood glucose control before and after the intervention period. Of these available results, there was an average decrease of 1.6% in the computer group and an average increase of 1.0% in the noncomputer group. However, nothing of significance can be said about these limited data.

Computer System Use

Data collected by the FirstClass software system included the mean number of minutes that a woman used the system each month (Figure 1). Across the 5-month period, the group averaged 63.8 minutes per month. The average time logged on in the first month was 129 minutes, with an average of 37 minutes in the month with the lowest use (month 5). There was a tendency for the women to spend more time using the computer during the first month, which may have been due to learning the system or having something new to use. Subsequent comments on the attitude survey showed that at least part of the variation was due to seasonal changes in their schedules.

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Table 2.

Supportive/Informational Conversation

Foot Care	Nutrition
Woman 1: Does anybody have a good cream for dry skin? My heels are horrible. . . . I have lots of trouble with calluses on my heels. They crack and then split open. . . . I go barefoot in the house and wear thongs outside. I know I should wear shoes, but they are so uncomfortable for me and my feet get so hot. So, has anybody run across any good lotion they feel really helps, and does anybody have the troubles I've had?	*The women were invited to post their questions about diet to a CDE dietitian who came online and addressed each woman's concerns individually. In addition, the women acted as resources for each other.*
Many suggestions were offered for a variety of lotions, but the most dramatic advice came in response to the fact she was not wearing shoes.	*Woman 1:* When it comes to diet, I was wondering if someone could work with me on how to make a menu of low-fat food that I like, which would be a real challenge because I am not a big fan on a lot of the fruits and veggies—just a few of them, but I don't know how to cook that way or I don't have recipes. I would like recipes that are down-to-earth, things that a person has in their house. Is anyone else interested in this?
Woman 2: I have seen feet that are like you mention. Get some Vaseline and then wear a pair of white anklets to bed and get some comfortable slippers for in the house or you'll have bad to worse—and when you get out of the tub or shower be sure your feet are dry. . . . Keep up the care of your feet and stop going barefoot—that is the most dangerous for a diabetic to injure your feet. Honest, please, get some canvas shoes or walking shoes—something—and wear slippers in the house!	*Woman 2:* I really don't have any recipes for low-fat, but I do have helpful hints. Bake your chicken on a rack, remove the skin. Bake a cake using applesauce instead of oil—use the same amount. Use Egg Beaters. Wash your hamburger after it is fried. I fry the meat, put it in a colander, let it drain into a can. Then I turn cold water on and run it over the meat and actually wash it.
Woman 3: Really keep an eye on those cracks. Any open sore, especially on the feet, is a real red flag for a diabetic. I never go barefoot, even in the house—and check your feet daily for sores or cracks. If you have any questions, ask your doctor.	*Woman 3:* I might be able to help you with the ordinary food and some ways of eating that will help you lose the weight. But you must realize and remember the loss will be slow and seem to take forever, but you won't gain, and you won't be hungry, and it is the most healthy diet in the world. I'll make a copy of it and send it to you. . . . You will love the monthly loss of 4 pounds . . . gone forever! There is also lots of good help in the big book you got with the WTW Project.

Data also were collected about the number of times each component of the software was used. The "Conversation" area, which was the support group component, was the most frequently used. This component allowed women to ask and respond to questions, talk about problems, and communicate in ways that fostered supportive help. Each woman used the "Conversation" area an average of 111 times over the course of the 5 months. The "Mailbox" area, where private conversations took place, was the next most often used feature, with each woman averaging 40 exchanges over the course of the computer intervention. Although the "Mailbox" component was not in a group format, it also allowed the women to give and receive supportive help. The "HealthChat" area was accessed an average of 38 times by each participant. This included the time spent by the women reading and responding to the CDE's self-management training and some nonstructured health discussions that were guided by the CDE.

Impact of the Computer Support Group

The effect of the computer-based support group on the lives of the women was expressed in a number of ways. Informational help, although provided by the CDE through the "HealthChat" area, was supplemented by the personal knowledge of the women themselves. Supportive help, which Weiss[1] defined as "attachment, being an integral part of a group, nurturant behavior, reassurance of worth, and emotional help," was woven throughout the women's exchanges with each other. Examples of conversation-type exchanges that demonstrate the interaction between support and information are shown in Table 2. As can be seen in the flow of conversation during the computer intervention, the project was perceived as a valuable means of offering the informational dimension of support, both from the women themselves and from the CDE.

Attitude Survey

The attitude survey, which was administered to the computer group in month 6, also reflected a positive response

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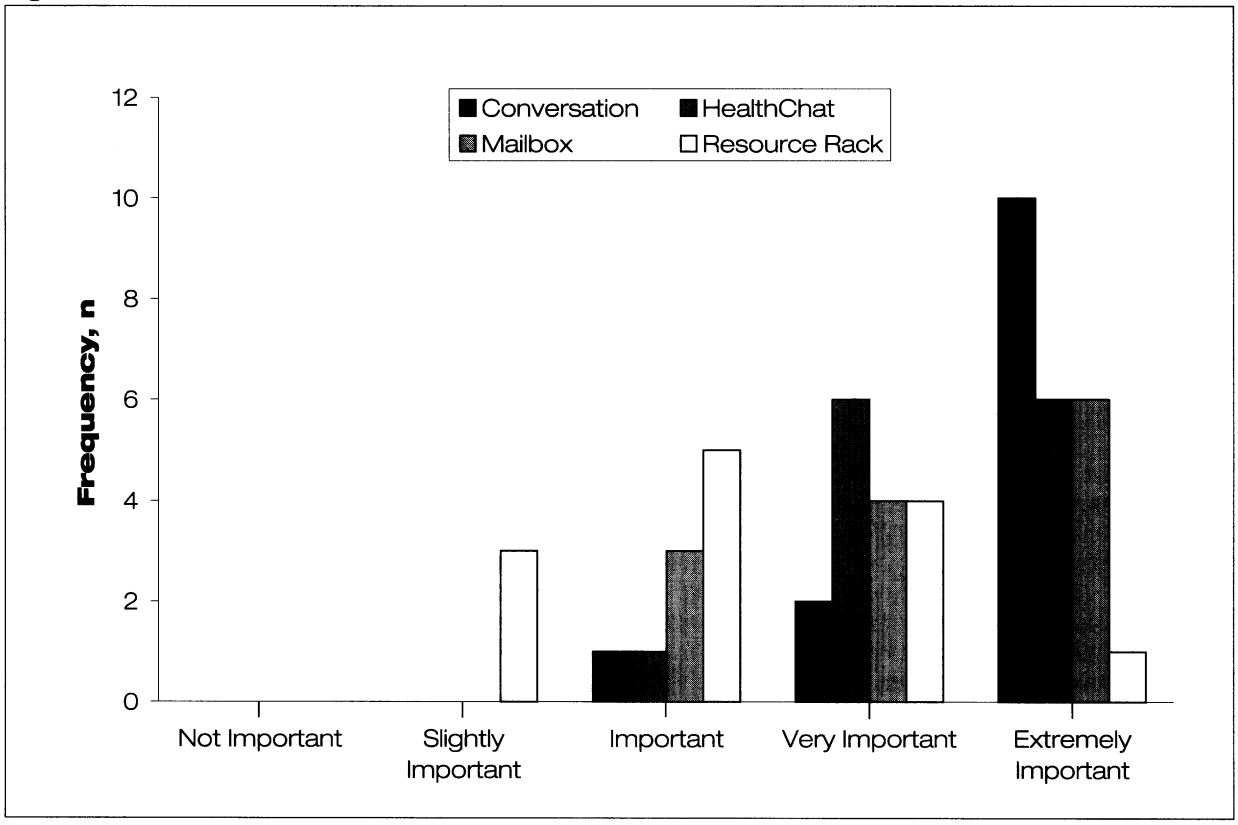
Figure 2.



Importance of components.

to both the support of other women in the group and telecommunication as an educational tool. Of the women who completed the project, 77% said the project provided a great deal of support for them; none said that it did not provide any support. Twelve women felt that the project gave them a significant sense of connectedness. The women were asked on the survey how important they considered each of the components of the FirstClass software system. Their responses are compared in Figure 2. The "Conversation" and "HealthChat" components were considered the most important. This finding was also reflected in the number of times that these 2 areas were accessed on the computer.

When asked, 77% of the women said they would very highly recommend the project to other women, and 23% said they would recommend the project to other women. None said that they would not recommend the project to other women.

The open-ended questions on the attitude survey gave the women an opportunity to express both negative and positive thoughts and feelings about the intervention. The following examples show the positive attitude and thoughtful suggestions about the project expressed by the women in the computer group in response to this open-ended question: If a reporter from CNN approached you to talk about being a participant in the Women to Women Diabetes Project, how would you describe your experience?

Woman 1: I very highly recommend the Women to Women Project for any women who have IDDM or NIDDM, no matter where they live because it allows you to share and learn when you have the time.

Woman 2: I was involved in the Women to Women Project aimed at providing rural women with a support group, information on their disease, resources for their disease, and friendships throughout the state to help in the control of their diabetes. We talked about problems, complications, concerns, insulin, diet, health matters, diabetes supplies . . . you name it, and we discussed it! The project made me realize that I am not alone in dealing with my diabetes and that it helps sometimes to share your feelings and concerns with others.

Woman 3: It was a unique experience where the distance was shrunk and you could talk about your problems with someone who really understood because they had gone through it themselves.

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Woman 4: [I] would relay how difficult it is in rural areas to have access to a diabetes specialist, re: travel time, travel miles (bad weather). I would always wonder if I was doing everything I possibly could to enhance the chance of little or no complications from IDDM and to ensure quality of life. The study was a wonderful opportunity to communicate with other women—compare notes, feelings, friendship, and receive quality reading material. Larger cities have support groups, have everything which we don't have access to. Communicating with the others kept me 'on track' and encouraged me to always do a little bit more, try harder, change my (our) lifestyle, and eat more healthy.

The following examples are responses to the other open-ended question: Is there anything else that you would like for us to know about your experience of participating in the Women to Women Diabetes Project? Any ideas about the program that you would suggest be changed as we continue with this project?

Woman 5: Summertime finds most of us short of time. When we first started in March, it seemed like we took more time for the program. Winter is dark and cold, and I think more women are depressed, and the support system really works. In summer the sun is out, life renews itself, we can work outside, and life seems better.

Woman 6: Great job. Please don't stop. There are more people out there that you all can help.

Woman 7: I would love to have some help setting up a support group here—maybe for different age groups—one for teens and younger, and adults.

Another measure of satisfaction with the support of other women with diabetes was their motivation to continue contact with others who participated in the intervention group. During the telephone interviews at 7½ months and 10 months, the women in the computer group were asked if they had continued contact with other women in the computer group. Seventy-seven percent said they had kept in touch with at least one woman who was part of the intervention group since the end of the computer intervention.

CONCLUSIONS

The number of participants that could be supported by this study was small, so definitive conclusions cannot be drawn. Even so, this study was a source of valuable insights and initial indications of the success of the computer intervention. It was demonstrated that computers are easily used and the software system can be accessed from isolated rural areas without substantial difficulty. The women clearly indicated that they had received support from each other and training from the CDE. Despite vacations, spring/summer agricultural responsibilities, and other distractions, they were able to access support at times convenient for them.

Replication with a larger population of women with diabetes would yield more opportunity to demonstrate statistical significance. Expanding the study to different populations (eg, men, Native Americans, etc) may reveal different support and training issues. Modifying the protocol of this project to include measures of HbA_{1c} by a method other than self-report, specific knowledge improvement measures, and motivation-for-change measures may further support the efficacy of this technology for use by CDEs in rural settings.

In addition to the specified aims of this project, this study served as the initial phase of a larger Women to Women Project being conducted at Montana State University College of Nursing and Burns Telecommunication Center. This ongoing study involves 120 isolated rural women with chronic diseases. The Women to Women Diabetes Project is the basis for the statistical modeling that is being used in the larger project.

The WTWDP also demonstrated that setting up computers in the homes of rural and isolated rural people was not only possible, but relatively uncomplicated. A toll-free help line was available for contacting the study team. After the first few weeks, calls for technical support were rare. One laptop computer had to be replaced due to hardware problems. The replacement computer was mailed to the participant and instructions for set up were given over the telephone. This was accomplished in 2 telephone conversations and no other problems were reported. Another technical difficulty was with the rural area telephone lines. When there was wind or the lines were very busy, the line static would disconnect the modems. Once the women were made aware of this problem, they would avoid using the system at times during which line static might be expected or would choose to log back on several times during a session.

IMPLICATIONS

The Women to Women Diabetes Project introduced another way to use evolving technology to improve diabetes self-management training. The use of this technology could save time and enhance client satisfaction by introducing the flexibility of asynchronous but interactive communication.

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In addition to improving the professional/client interaction, there is a potential for introducing or enhancing client-to-client interaction. The CDE may want to seriously consider such a plan in light of the importance of social support to people managing a chronic illness.

During this time of economic challenges for the CDE, this technology could also provide a cost-effective method of delivering quality diabetes self-management training.[23] The cost of the professional's telephone time could be decreased, and the ability to reach a larger number of people with diabetes could be increased. The professional could use this computer telecommunication style of support to interact with more than one person at a time using a classroom style of teaching. The class would be taught at the professional's convenience and attended at the client's convenience. The students in the online classroom could also interact with each other as their schedules permitted.

Managing a client's blood glucose would be enhanced by using the "Mailbox" component of the system to discuss strategies for individualized management. Again, the conversation would take place at the convenience of both the CDE and the client. Other software could be incorporated into the practice of the CDE, such as the downloading of self-monitoring of blood glucose data from the memory of home glucose monitors and the electronic transfer of patient education materials. In this way, an integrated record-keeping/client-teaching/support-group system would become feasible.

Telecommunication is a viable option for enhancing access of rural women to social support and diabetes self-management training systems. This creative approach has been well received by women living in rural areas and is highly valued by those involved in its use.

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